

Laboratório 3: Dados relacionais e operações do tipo join

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```
library(readr)
library(dplyr)
library(leaflet)
```

1. Importação dos dados

1.1 Aeroportos

Importando só as colunas pedidas: IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE.

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)

airports
```

```
# A tibble: 322 x 5
  IATA_CODE CITY      STATE LATITUDE LONGITUDE
  <chr>      <chr>      <chr>   <dbl>   <dbl>
1 ABE      Allentown    PA      40.7    -75.4
2 ABI      Abilene      TX      32.4    -99.7
3 ABQ      Albuquerque NM       35.0   -107.
4 ABR      Aberdeen    SD      45.4    -98.4
5 ABY      Albany      GA      31.5    -84.2
6 ACK      Nantucket    MA      41.3    -70.1
7 ACT      Waco         TX      31.6    -97.2
8 ACV      Arcata/Eureka CA       41.0   -124.
```

```

 9 ACY      Atlantic City NJ      39.5      -74.6
10 ADK      Adak      AK      51.9      -177.
# i 312 more rows

```

1.2 Companhias aéreas

```

airlines <- read_csv("airlines.csv")

airlines

```

```

# A tibble: 14 x 2
  IATA_CODE AIRLINE
  <chr>      <chr>
1 UA        United Air Lines Inc.
2 AA        American Airlines Inc.
3 US        US Airways Inc.
4 F9        Frontier Airlines Inc.
5 B6        JetBlue Airways
6 OO        Skywest Airlines Inc.
7 AS        Alaska Airlines Inc.
8 NK        Spirit Air Lines
9 WN        Southwest Airlines Co.
10 DL       Delta Air Lines Inc.
11 EV       Atlantic Southeast Airlines
12 HA       Hawaiian Airlines Inc.
13 MQ       American Eagle Airlines Inc.
14 VX       Virgin America

```

1.3 Vôos (leitura em partes / *chunks*)

Lendo `flights.csv` em partes de 1 milhão de linhas, aplicando os filtros e calculando as estatísticas suficientes em cada parte.

```

#| message: false

juntar_chunk <- function(data, index, acc) {

  data |>
    filter(!startsWith(DESTINATION_AIRPORT, "1")) |>
    tidyr::drop_na() |>

```

```

    group_by(DESTINATION_AIRPORT) |>
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      n_voos      = n(),
      .groups     = "drop"
    )
  }

estatisticas <- read_csv_chunked(
  file       = "flights.csv",
  callback   = AccumulateCallback$new(juntar_chunk),
  chunk_size = 1e6,
  col_types  = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY       = col_double()
  )
)

# junta as partes e calcula o atraso medio final
flights <- estatisticas |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    soma_atraso = sum(soma_atraso),
    n_voos      = sum(n_voos),
    .groups     = "drop"
  ) |>
  mutate(ARRIVAL_DELAY_MEDIO = soma_atraso / n_voos) |>
  select(DESTINATION_AIRPORT, ARRIVAL_DELAY_MEDIO, n_voos)

flights

```

A tibble: 308 x 3

	DESTINATION_AIRPORT	ARRIVAL_DELAY_MEDIO	n_voos
	<chr>	<dbl>	<int>
1	ABE	4.23	313
2	ABI	0.252	330
3	ABQ	6.64	2753
4	ABR	0.0577	104
5	ABY	8.88	133
6	ACT	0.391	253
7	ACV	13.3	210
8	ACY	3.98	509

```
9 ADK -2.7 10
10 ADQ -2.43 47
# i 298 more rows
```

2. Operação *join*

2.1 Chave em flights

A chave é DESTINATION_AIRPORT.

2.2 Chave em airports

A chave é IATA_CODE.

2.3 Aeroportos em flights ausentes em airports

```
aeroportos_ausentes <- anti_join(
  flights, airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)

aeroportos_ausentes
```

```
# A tibble: 0 x 3
# i 3 variables: DESTINATION_AIRPORT <chr>, ARRIVAL_DELAY_MEDIO <dbl>,
#   n_voos <int>
```

2.4 Comando de combinação (join)

Quero manter todos os aeroportos de `flights` e só adicionar as colunas de `airports`. Por isso uso `left_join`, com `DESTINATION_AIRPORT` e `IATA_CODE` como chaves.

```
flights <- left_join(
  flights, airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
```

2.5 Tabela resultante

```
flights
```

```
# A tibble: 308 x 7
```

	DESTINATION_AIRPORT	ARRIVAL_DELAY_MEDIO	n_voos	CITY	STATE	LATITUDE	LONGITUDE
	<chr>	<dbl>	<int>	<chr>	<chr>	<dbl>	<dbl>
1	ABE	4.23	313	Alle~	PA	40.7	-
75.4							
2	ABI	0.252	330	Abil~	TX	32.4	-
99.7							
3	ABQ	6.64	2753	Albu~	NM	35.0	-
107.							
4	ABR	0.0577	104	Aber~	SD	45.4	-
98.4							
5	ABY	8.88	133	Alba~	GA	31.5	-
84.2							
6	ACT	0.391	253	Waco	TX	31.6	-
97.2							
7	ACV	13.3	210	Arca~	CA	41.0	-
124.							
8	ACY	3.98	509	Atla~	NJ	39.5	-
74.6							
9	ADK	-2.7	10	Adak	AK	51.9	-
177.							
10	ADQ	-2.43	47	Kodi~	AK	57.7	-
152.							

```
# i 298 more rows
```

3. Aeroportos por estado

```
aeroportos_por_estado <- airports |>
  group_by(STATE) |>
  summarise(n_aeroportos = n(), .groups = "drop") |>
  arrange(desc(n_aeroportos))

aeroportos_por_estado
```

```
# A tibble: 54 x 2
```

```

      STATE n_aeroportos
    <chr>      <int>
1 TX          24
2 CA          22
3 AK          19
4 FL          17
5 MI          15
6 NY          14
7 CO          10
8 MN           8
9 MT           8
10 NC          8
# i 44 more rows

```

4. Mapa dos atrasos observados por aeroporto

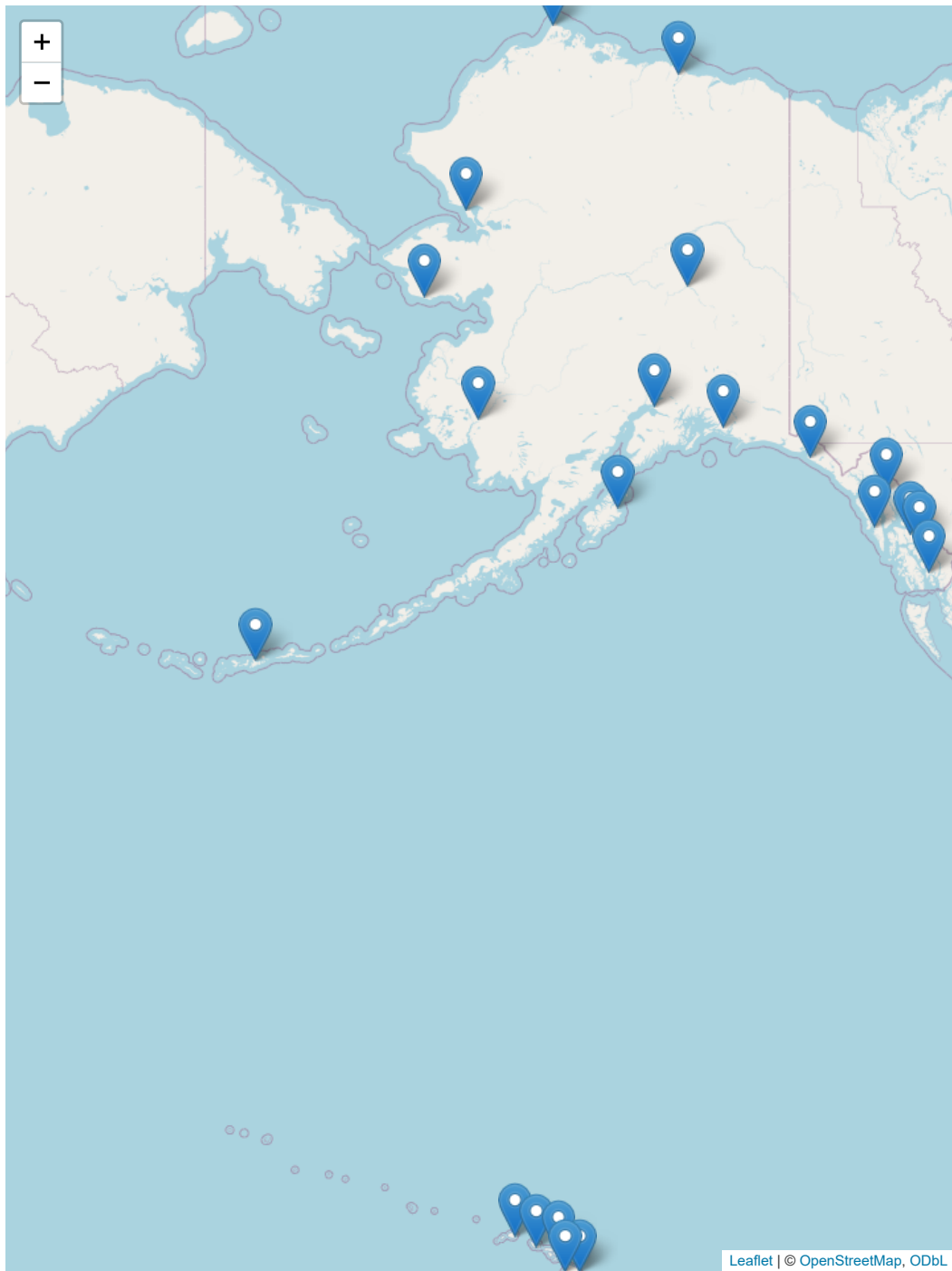
```

this <- leaflet(data = flights) |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      DESTINATION_AIRPORT, " (", CITY, ", ", STATE, ")<br>",
      "Atraso médio: ", round(ARRIVAL_DELAY_MEDIO, 2), " min"
    )
  )

```

Warning in validateCoords(lng, lat, funcName): Data contains 3 rows with either missing or invalid lat/lon values and will be ignored

```
this
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 02:31:18 -03"
```